Topic 21 — World Minerals & Energy

Covers syllabus (click to expand)

Coalfields | Iron ore | Copper | Tin | Petroleum / OPEC | Natural gas | Conventional vs renewable energy (नवीकरणीय ऊर्जा) | Other minerals (Mn, bauxite, gold, diamond, U, phosphate) | Critical minerals (REE, lithium, cobalt)

Sources baked in: Lucent world mineral-field match-lists; NCERT Class 12 (minerals & energy); coaching resource map (LotusArise / shields-basins); **Ghatnachakra** Minerals (CB-223+); UPPCS Prelims 2018-2025 **Weight:** ★★★★ — coalfield-country, Mount Newman iron, Chile copper, Kinta tin, Kashagan ≠ Kuwait, Pegu Yoma oil; teach full field map **Last verified:** August 2026 — coaching gap-pass (Lithium Triangle, REE-China, Carajás, producer≠exporter) **Current Affairs:** none mandatory for Prelims field matches (producer ranks freeze to the PYQ year)

Current Affairs (this topic)

None mandatory. Producer ranks and "leading country" freeze to the **paper's year**. Do not chase every USGS revision in the teaching home.

UKPCS Prelims 2025

Q. UKPCS Prelims 2025, Q83

Which country is the leading producer of gold in the world in the year 2023?

A. China B. U.S.A. C. Canada D. South Africa

Show answer

Logic: South Africa is the historic (ऐतिहासिक) leader, not the 2023 leader. **Ans: A (Series B provisional key).** China leads recent world gold output; South Africa is the twentieth-century trap.

N.0 Framework — World Minerals & Energy

Identity: This chapter is a field-country / mineral-place match set. Geology comes first, then the standard field names.

Precambrian shields host many metallic ores.

Sedimentary basins host coal, oil and gas.

Andean porphyry belts host copper, with lithium brines nearby in the same broad story.

Fact **one identity line** per famous field.

Learn mineral, country and one trap.

Separate **producer volume** from **exporter**.

China often mines iron and coal and still imports them.

India mineral belts live mainly in the Indian Minerals chapter.

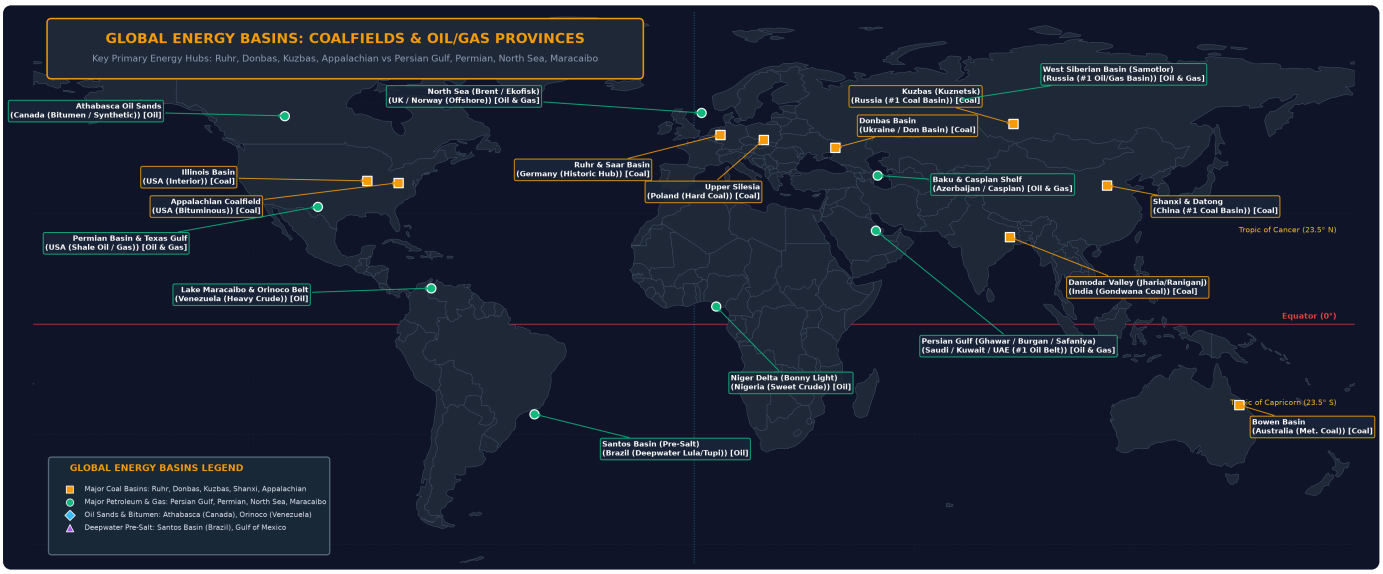
Here keep only world centres that sneak into papers, plus a few India tags already in PYQs.

Order of this chapter

1. Coal → iron → copper → tin.
2. Petroleum / OPEC → natural gas → energy types.
3. Other minerals + critical minerals (REE, lithium, cobalt).

Teach the **full field map**. Next year can ask any Lucent coalfield or ore town, not only Ruhr or Kinta.

N.1 Coal



- Fossil fuel from buried peat in swamp forests (Carboniferous in Europe–USA; **Gondwana** in India–Australia–South Africa).
- Used as **steam coal** (thermal power) and **coking coal** (steel). Coking ≠ lignite.
- Ranks **rise in carbon** and fall in moisture: peat → lignite → bituminous → anthracite.

Types

Type	Carbon	Character	Region
Peat	Lowest	First stage; high water	Ireland, Russia, Canada bogs
Lignite (brown coal)	Low	Soft; thermal only	Germany (Lusatia), Australia, Neyveli-type
Bituminous	Medium–high	Steam **and** coking	Ruhr, Appalachian, Damodar (दामोदर), Kuzbass

Type	Carbon	Character	Region
Anthracite	Highest	Hardest; cleanest burn	**Pennsylvania** (USA); South Wales

Producer vs exporter (ranks move; freeze a year only if quoted)

Role	Usual names
Volume miners	**China** (eats most of it), India, Indonesia, USA, Australia, Russia, South Africa
Exporters	**Indonesia, Australia, Russia, USA, South Africa**
Trap	China is **not** the classic export king — it is the volume **consumer**

World coalfields — country fact

Coalfield	Country	Note
Appalachian	**USA**	Eastern USA (not the Rockies as the name)
Pennsylvania anthracite	USA	Highest-rank US coal
Interior / Illinois	USA	Bituminous
Lancashire	**England**	Historic English coal–textile belt
Yorkshire / Northumberland / Midlands	England	Same island; do not place on the USA
South Wales	UK	anthracite
Ruhr	**Germany**	**Not** Britain / France / Russia
Saar	Germany–France border	Also revise
Lusatia / Rhineland	Germany	**Lignite**
Silesia (Upper)	**Poland**	
Kuzbass / Kuznetsk	**Russia**	Siberia
Pechora	Russia	Arctic
Kansk–Achinsk	Russia	Lignite / brown coal
Donets (Russian side)	Russia	Do not swap with Ukraine Donetsk in a “country” MCQ
Donetsk / Donbas	**Ukraine**	2023 largest among Kyiv / Kharkiv / Zaporizhzhia
Karaganda	Kazakhstan	Coal city; 2020 iron list also tagged it
Shanxi / Shaanxi / Inner Mongolia	**China**	Volume belt
Hunter Valley (NSW)	Australia	Export

Coalfield	Country	Note
Bowen (Queensland)	Australia	Export
Witbank / Highveld	South Africa	
Damodar — Jharia, Raniganj	India	Jharia = coking; Raniganj = West Bengal

The 2025 coal matchs Appalachian to the **USA**, Lancashire to **England**, Ruhr to **Germany**, and Kuzbass to **Russia**. Ukraine's coal centre among the usual city options is **Donetsk**.

Inline PYQ — UPPCS Prelims 2025, Q4

Match List-I with List-II and choose the correct answer by using the code given below the lists.

List-I (Coal Field)	List-II (Country)
A. Appalachian	1. England
B. Lancashire	2. Germany
C. Ruhr	3. Russia
D. Kuzbass	4. USA

Options:

A. 4 1 2 3

B. 1 4 3 2

C. 1 4 2 3

D. 4 1 3 2

Show answer

Ans: A — Appalachian–USA, Lancashire–England, Ruhr–Germany, Kuzbass–Russia.

Inline PYQ — UPPCS Prelims 2023, Q69

Which one of the following is the largest coal-producing centre of Ukraine?

Options:

A. Zaporizhzhia

B. Kharkiv

C. Donetsk

D. Kyiv

Show answer

Ans: C — Donbas / Donetsk coal. Kyiv is the capital, not the coal answer.

Inline PYQ — UPPCS Prelims 2022, Q112

The famous **'Ruhr coalfield'** is situated in which of the following countries?

Options:

A. Great Britain

B. Germany

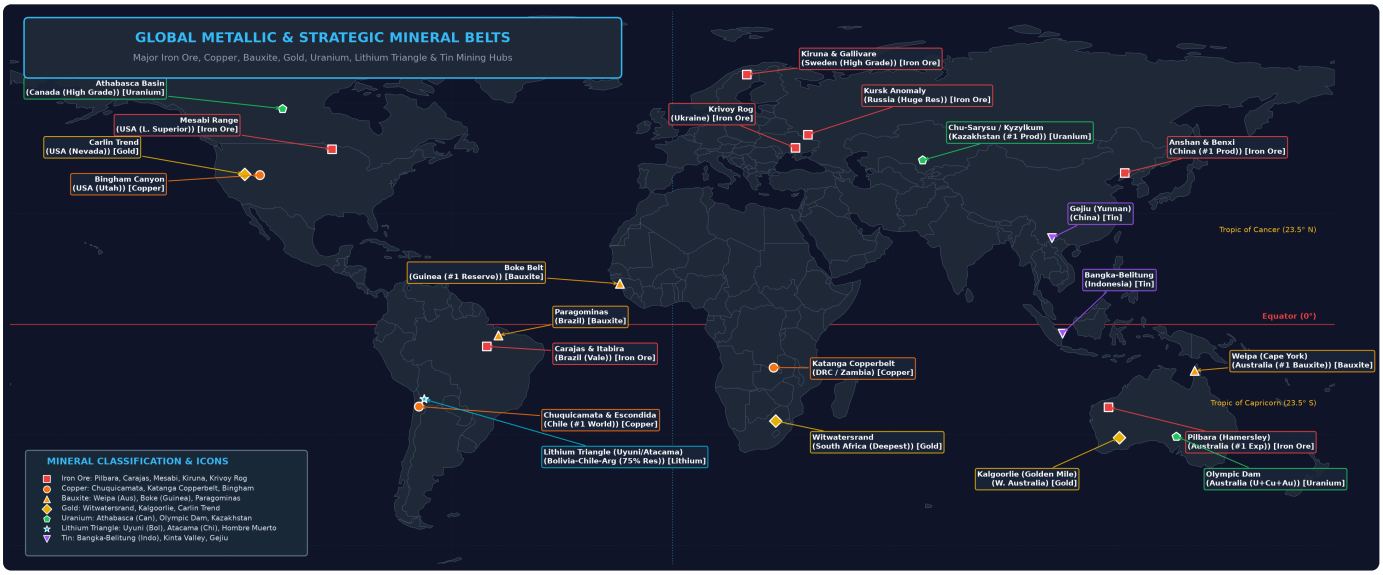
C. Russia

D. France

Show answer

Ans: B

N.2 Iron ore



Why it is asked

- Backbone of **steel** and heavy industry.
- Steel is iron + carbon (plus alloys). Pig iron from the blast furnace; steel from the converter.

Ores

Ore	Colour	Fe	Fact
Hematite	**Red**	~70%	**Bulk** ore of world trade (पण्याध्यक्ष)
Magnetite	**Black**	~72%	**Highest grade**; magnetic
Limonite	Yellow-brown	~60%	Hydrated; lower grade
Siderite	Grey-brown	~40%	Iron carbonate; poorest of the four (चातुर्यम)

Occurrence

- Banded iron formations (sedimentary) is Australia Pilbara, Brazil, India, Ukraine.
- Magmatic magnetite is Kiruna (Sweden).
- Needs coking coal + limestone nearby for a cheap steel town (Ruhr, Appalachian–Pittsburgh, Damodar–Jamshedpur story).

Producer vs exporter

Role	Usual names
Volume miners	Australia, Brazil, **China**, India, Russia
Exporters	**Australia** (Pilbara) and **Brazil** (Minas / Carajás)
Trap	China mines a lot and still **imports**. India is a miner, not the Pilbara-style ship king

Australia (2023 name)

- State is **Western Australia**.
- Region is **Pilbara**.
- Range is **Hamersley**.
- Mine in the paper is **Mount Newman**.
- Mineral is **iron ore** — not manganese, copper, or bauxite.

World iron areas

Area	Country	Note
Mount Newman	**Australia**	2023. Pilbara / Hamersley
Hamersley Range	Australia	Same belt
Itabira	**Brazil**	Minas Gerais; old “iron quadrangle”
Carajás	Brazil	Giant Amazon-side mine
Cerro Bolívar	Venezuela	Also revise
Mesabi Range	**USA**	Lake Superior / Minnesota
Labrador Trough	Canada	Quebec–Labrador
Lorraine	**France**	Minette ore; **not** Germany
Normandy	**France**	**Not Germany** (2020 wrong pair)
Pyrenees	France	2020 accepted pair
Kiruna / Gällivare	**Sweden**	Magnetite; Arctic rail to Narvik
Bilbao / Basque	Spain	
Salzgitter / Siegerland	**Germany**	Real German iron — do **not** write Normandy

Area	Country	Note
Cleveland / Scunthorpe	UK	Historical
Krivoy Rog / Kryvyi Rih	**Ukraine**	2020 correct pair. **Not** Russia
Kursk Magnetic Anomaly	Russia	
Magnitogorsk / Urals	Russia	Steel town on the ore
Karaganda	Kazakhstan	Coal city; 2020 iron list treated as matched
Anshan / Wuhan	China	
Sishen	South Africa	
Liberia / Mauritania	W Africa	Classic export names
Odisha–Chhattisgarh–Jharkhand–Karnataka	India	Bailadila, Noamundi, Kudremukh — India-map detail

Steel tonnage

- Early **2022** paper: **China** #1.
- India and Japan are large — **not** England as world top.

Mount Newman is an **iron ore** mine in the Pilbara. The Germany–Normandy iron pair is **wrong** — Normandy is in **France**.

Inline PYQ — UPPCS Prelims 2023, Q70

"Mount Newman" is famous for which of the following minerals?

Options:

A. Iron Ore

B. Manganese

C. Copper

D. Bauxite

Show answer

Ans: A — Pilbara, Western Australia.

Inline PYQ — UPPCS Prelims 2020, Q70

Which one of the following is NOT correctly matched?

(Country) (Iron-Ore Producing Area)

Options:

A. Kazakhstan – Karaganda

B. Ukrain – Krivoy Rog

C. Germany – Normendy

D. France – Pyrenees

Show answer

Ans: C — Normandy is in **France**, not Germany. Krivoy Rog = Ukraine iron. Paper treated A and D as matched.

Inline PYQ — UPPCS Prelims 2022, Q140

As of early 2022, which country was at the top in steel production in the world?

Options:

A. China

B. Japan

C. England

D. India

Show answer

Ans: A — Freeze to early-2022. India is a large producer, not the 2022 top.

N.3 Copper

Properties / uses

- Best electrical conductor after silver → **wires**.
- **Brass** is copper plus zinc.
- **Bronze** (कांस्य) is copper plus tin.
- Copper is malleable and used in roofing, pipes, coins (रूपक) and electronics.

Ore

- The chief ore is **chalcopyrite** ($CuFeS_2$).
- Also revise chalcocite, bornite, malachite and azurite.

Porphyry copper (2022 Reason)

- Large, low-grade, **disseminated** Cu in volcanic-arc rocks.
- Northern **Andes of Chile** hold the 's "largest porphyry deposits".
- That is why **R explains A** in 2022: Chile leads **because** of those Andes porphyries.
- Rank vs DRC can flip in later USGS years — freeze **Chile + N Andes** to the 2022 paper.

World copper map

Place	Country	Note
Chuquicamata	**Chile**	Open-pit giant; Atacama / N Chile
El Teniente	Chile	One of the world's largest underground Cu mines
Escondida	Chile	Modern volume name
Andes (Peru)	Peru	2nd Andes producer
Katanga (Shaba)	**DRC**	African copper-cobalt
Copperbelt	**Zambia**	Ndola–Kitwe belt
Bingham Canyon	**USA**	Utah
Butte	USA	Montana; historical
Sudbury	Canada	**Nickel + copper** together
Norilsk	Russia	Ni–Cu–PGE
Grasberg	Indonesia	Cu + gold (Papua)
China	China	Huge **user**; also mines
Khetri / Malanjkhand / Singhbhum	India	India-map fact

Producer fact (2022)

- **Chile** is leading producer in that paper.
- Do not write "Chile has no copper".

Inline PYQ — UPPCS Prelims 2022, Q41

Given below are two statements, one is labelled as Assertion (A) and the other as Reason (R).

Assertion (A): Chile is the leading producer of copper in the world.

Reason (R): The Andes Region of North Chile is enriched with the largest deposits of porphyry copper.

Options:

A. (A) is false but (R) is true.

B. (A) is true but (R) is false.

C. Both (A) and (R) are true but (R) is not the correct explanation of (A).

D. Both (A) and (R) are true and (R) is the correct explanation of (A).

Show answer

Ans: D — Learn **Chile + northern Andes porphyry**. Later-year DRC rank fights do not rewrite the 2022 key.

N.4 Tin

Ore / uses

- The ore is **cassiterite** (SnO_2). It occurs in alluvial (जलोढ़) placers and granite lodes.
- Uses include **tin plate** (cans), solder, **bronze** (Cu + Sn) and pewter.
- Tin is a soft, corrosion-resistant coating on steel.

SE-Asia tin belt (the world tin street)

- The belt runs **Myanmar (Tenasserim)** → **Thailand** → **Malaysia** → **Indonesia**.
- Malaysia's classic name is **Kinta Valley** (Perak) — **tin**, not rubber, tea or coffee (2020).
- Indonesia's classic names are the **Bangka and Belitung** islands.
- Myanmar tin sits in **Tenasserim / Dawei / Mawchi** — **not** the Pegu Yoma.

Other tin countries

Place	Country	Note
Kinta Valley	**Malaysia**	2020 PYQ. Rubber grows in Malaysia; Kinta still = **tin**
Bangka / Belitung	Indonesia	
Yunnan / Guangxi	China	Usual **volume** leader in modern lists
Potosí / Andes lodes	**Bolivia**	Highland tin
Andes	Peru	
Jos Plateau	Nigeria	Older textbook name
Cornwall	UK	Historical; worked out
Rondônia (राष्ट्रीय अन्वेषण एजेंसी)	Brazil	Also revise

Pegu Yoma is oil, not tin

- The **Pegu Yoma (Bago Yoma)** are the central Myanmar hills between the Irrawaddy and the Sittaung.
- The 2022 important mineral for this name is **mineral oil / petroleum**.
- Yenangyaung–Chauk oilfields sit in this central belt.
- Trap: Myanmar *does* mine tin — just **not under the Pegu Yoma label**.

Inline PYQ — UPPCS Prelims 2020, Q88

In Malaysia, the 'Kinta Valley' is famous for

Options:

A. Rubber Production

B. Tea Production

C. Tin Production

D. Coffee Production

Show answer

Ans: C

Inline PYQ — UPPCS Prelims 2022, Q70

Which among the following is an important mineral of the Pegu Yoma region of Myanmar?

Options:

A. Mineral oil

B. Tin

C. Copper

D. Silver

Show answer

Ans: A — Mineral oil. Tin is the SE-Asia-belt trap (Kinta / Bangka / Tenasserim), not this PYQ's Pegu Yoma tag.

N.5 Petroleum

What it is

- Petroleum is liquid hydrocarbon in **sedimentary** basins.
- The classic trap structure is an **anticline** (also fault, salt (लवणाध्यक्ष) dome, offshore shelf).
- Associated gas often sits above the oil.

OPEC

- Headquarters is **Vienna**.
- The original five (1960) were **Iran, Iraq, Kuwait, Saudi Arabia and Venezuela**.
- The Gulf core still dominates reserves. Indonesia has been in and out. Qatar left. Russia is **OPEC+**, not OPEC.

Producer vs reserve trap

- **Reserves** names: Venezuela, Saudi Arabia, Iran, Canada (oil sands), Iraq.
- **Production** names: USA, Saudi Arabia, Russia (ranks move).
- Freeze a year only if the paper quotes it.

World oil-field match-list

Field / area	Country	Trap
Ghawar	Saudi Arabia	Often tagged world's largest conventional field

Field / area	Country	Trap
Dhahran	**Saudi Arabia**	2024 correct pair
Abqaiq / Safaniya	Saudi Arabia	Safaniya = offshore
Ras Tanura	Saudi Arabia	Export **terminal**, not a field name swap
Burgan	**Kuwait**	Kuwait's giant. **Not** Kashagan
Kirkuk / Mosul	**Iraq**	North Iraq
Rumaila / **Zubair** / Basra	Iraq	South Iraq. Zubair = 2024 correct
Haft Kel	**Iran**	2024 correct. **Not** Iraq
Masjid-i-Suleiman	Iran	First Middle-East commercial well (1908 story)
Agha Jari / Gachsaran	Iran	
Kashagan	**Kazakhstan**	Caspian offshore. **Kuwait–Kashagan = WRONG** (2024)
Tengiz	Kazakhstan	Onshore Caspian
Baku	Azerbaijan	Caspian; Nobel/old Russian oil story
West Siberia / **Samotlor**	Russia	Volume belt
Volga–Urals	Russia	Older Russian belt
North Sea	**UK + Norway**	Ekofisk (NO), Forties/Brent (UK)
Prudhoe Bay	USA (Alaska)	
East Texas / Permian / Gulf of Mexico	USA	
Maracaibo / Orinoco	Venezuela	Heavy oil / sands story
Cantarell	Mexico	
Campos / pre-salt	Brazil	
Hassi Messaoud	Algeria	
Bonny / Niger Delta (डेल्टा)	Nigeria	
Dukhan	Qatar	
Murban	UAE	
Daqing	China	
Minas	Indonesia	
Seria	Brunei	
Yenangyaung / Chauk	**Myanmar**	With **Pegu Yoma** oil (2022)
Mumbai High / Digboi / Ankleshwar	India	India-map fact

Kuwait–Kashagan is **wrong**. Kashagan sits in **Kazakhstan** (Caspian). Kuwait's classic giant is **Burgan**.

Inline PYQ — UPPCS Prelims 2024, Q78

Which one of the following pairs (Country - Oil Field) is not correctly matched?

Options:

A. Iran - Haft Kel

B. Kuwait - Kashagan

C. Saudi Arabia - Dhahran

D. Iraq - Zubair

Show answer

Ans: B — Kashagan is in **Kazakhstan**. Kuwait's classic giant is **Burgan**.

N.6 Natural gas

Composition (PYQ fact)

- Main constituent is **methane (CH_4)** — used in 2018 Q130 and 2022 Q19.
- The same methane fact covers **CNG**.
- **LPG** is propane plus butane. LPG is not the same as CNG.
- **LNG** is methane cooled to liquid for **ship** trade (Qatar, Australia).

Types

- **Associated** gas comes with oil in the same field.
- **Non-associated** gas comes from gas-only fields (Qatar North Field / Iran South Pars, Groningen, Urengoy).

Producer / reserve names (ranks move)

Role	Usual names
Reserves	**Russia, Iran, Qatar**, USA, Turkmenistan
Production	**USA, Russia**, Iran, China, Qatar, Australia, Norway
LNG export	**Qatar, Australia**, USA

World gas fields

Field / area	Country	Note
Urengoy / Yamburg / Medvezhye	Russia	West Siberia
North Field	**Qatar**	Same giant reservoir as Iran's South Pars

Field / area	Country	Note
South Pars	**Iran**	Persian (फ़ारसी) Gulf; world's largest gas accumulation with Qatar
Groningen	Netherlands	Historic European giant
Troll / Ekofisk gas	Norway	North Sea
Hugoton / Marcellus shale	USA	
Hassi R'Mel	Algeria	
Galkynysh	Turkmenistan	
Shah Deniz	Azerbaijan	Caspian; pipelines to Turkey/EU
NW Shelf / Gorgon	Australia	LNG
North Sea	UK + Norway	Oil **and** gas
KG / Mumbai High gas	India	India-map fact

Inline PYQ — UPPCS Prelims 2022, Q19

Which among the following compounds is the main constituent of natural gas?

Options:

A. Butane

B. Hexane

C. Benzene

D. Methane

Show answer

Ans: D

Inline PYQ — UPPCS Prelims 2018, Q130

Which of the following is the main component of the natural gas?

Options:

A. Ethane

B. Butane

C. Carbon mono-oxide

D. Methane

Show answer

Ans: D

N.7 Energy resources

Conventional

- Coal
- Petroleum
- Natural gas
- Hydroelectricity
- Nuclear

Non-conventional

- Solar
- Wind
- Tidal
- Geothermal
- Biogas (बायोगैस)

Non-renewable: coal, petroleum, natural gas, nuclear.

Renewable: solar, wind, hydro, tidal, geothermal, biomass.

From the Sun

- Coal, oil, gas
- Biomass
- Wind, waves, hydro
- Direct solar

Not from the Sun

- **Nuclear** — radioactive minerals
- **Geothermal** — Earth's internal heat
- **Tidal** — Moon's gravity

Nuclear energy and geothermal energy are **not** stored solar. Wind and biomass **are** solar-linked in the usual Prelims logic.

Hydroelectricity

- Needs height of fall + volume of water.

Project	River	Country
Three Gorges	Yangtze	**China**
Itaipu	Paraná	**Brazil–Paraguay**
Grand Coulee	Columbia	USA
Hoover	Colorado	USA

Project	River	Country
Aswan High	Nile	Egypt (मिस्र)
Guri	Caroní	Venezuela
Churchill Falls	Churchill	Canada
Sayano-Shushenskaya	Yenisei	Russia

Nuclear

- The main nuclear fuel is **uranium**.
- Leading miners are **Kazakhstan**, Canada and Australia.
- The highest **share** of electricity from nuclear power is in **France**.
- The USA holds large **capacity**.
- The IAEA headquarters is in **Vienna**.

Solar and wind

- Largest capacity: **China**.
- High wind share: Denmark, UK.

Geothermal

Country	Fact
Iceland	Power + heating
New Zealand	Rotorua / Taupo
Philippines	Ring of Fire
USA	The Geysers, California
Italy	Larderello (first plant)

Tidal

- **La Rance** — France
- Sihwa — Korea
- India potential: **Gulf of Khambhat > KutchAgencies**

Body	HQ
OPEC	Vienna
IAEA	Vienna
IEA	Paris
IRENA	Abu Dhabi

N.8 Other world minerals

Manganese (steel alloy)

Place	Country
Postmasburg / Kalahari	**South Africa** (2020 — not U / mica / bauxite)
Moanda	Gabon
Groote Eylandt	Australia
Nikopol	Ukraine
Odisha / Maharashtra	India

Bauxite → **aluminium** (laterite (लेटराइट) ore; needs huge power to smelt)

Place	Country
Weipa / Gove / Darling Range	**Australia**
—	**Guinea** (reserves renown)
Jamaica / Suriname	Caribbean-Guiana
Trombetas	Brazil
Les Baux	France (name origin)
Odisha-Gujarat-Jharkhand	India

- Mount Newman is **iron**, not bauxite.

Gold

Place	Country
Witwatersrand	South Africa
Kalgoorlie	Australia
Yukon / Klondike	Canada
Homestake	USA (historical)
Grasberg	Indonesia (with copper)
—	China (modern volume)

Diamond

- Kimberley / Kimberlite story is **South Africa**.
- Volume names: Russia (Yakutia), Botswana, Canada, DRC.
- Golconda (गोलकोंडा) is historical **India**.

Nickel

- **Sudbury** (Canada) yields nickel with copper.
- **Norilsk** (Russia) is another Ni–Cu name.
- Also revise New Caledonia and Cuba (laterite Ni).

Chromite

- Bushveld (**South Africa**). Great Dyke (Zimbabwe). Kazakhstan. Turkey. Sukinda (India).

Lead–zinc

- **Broken Hill** and Mount Isa are Australian names.
- Sullivan is in Canada.
- Zawar and Rampura-Agucha are in India (Rajasthan).

Phosphate / potash (fertiliser)

- Phosphate rock: **Morocco** (world reserve king), Florida (USA), Kola (Russia), Jordan.
- Potash: Canada (Saskatchewan), Belarus, Germany, Dead Sea (Israel/Jordan).

Uranium / thorium

- Uranium mines: **Kazakhstan**, Canada Athabasca, Australia.
- Olympic Dam (ओलंपिक डैम, Australia) is U + Cu + Au.
- Thorium / monazite sands: India (Kerala–TN), Brazil, Australia.
- Postmasburg is **not** uranium.

Iodine / nitrate (Chile cluster)

- The **Iodine** 2018 fact (paper spelling **Iodine**) points to **Chile**.
- The source is Atacama **caliche** (also sodium nitrate).
- Do not mark Japan / USA / China as the 2018 leader.

Mica / graphite / sulphur / platinum / cobalt / lithium / tungsten

Mineral	World centre
Sheet **mica**	**India** (Jharkhand–Andhra–Rajasthan) classic
Graphite	China; India; Sri Lanka (vein)
Sulphur	USA, Mexico, Poland; volcanic Italy/Japan
Platinum	Bushveld (South Africa); Norilsk
Cobalt	**DRC** (with Katanga copper)
Lithium	Australia hard-rock (spodumene); **Lithium Triangle** brines = **Chile–Argentina–Bolivia** (Brazil is **not** in the triangle)
Rare earths (REE)	**China** dominates mining **and** especially **processing**; India has monazite sands but not China's refining fact
Tungsten	**China**; Bolivia

Mineral	World centre
Mercury	Almadén (Spain) historical
Asbestos	Russia, Kazakhstan; Canada historical

Inline PYQ — UPPCS Prelims 2020, Q84

Postmasburg and its adjacent areas of South Africa is major producer of which of the following minerals?

Options:

A. Uranium

B. Bauxite

C. Manganese

D. Mica

Show answer

Ans: C

Inline PYQ — UPPCS Prelims 2018, Q109

Which of the following countries is the leading producer of Iodine in the world?

Options:

A. Japan

B. U.S.A.

C. Chile

D. China

Show answer

Ans: C — Paper spelling **Iodine** = iodine. Chile Atacama.

Inline PYQ — UPPCS Prelims 2022, Q139

Most of the sources of energy we use represent stored solar energy. Which of the following is NOT ultimately derived from the Sun's energy?

Options:

A. Biomass energy (बायोमास ऊर्जा)

B. Nuclear energy

C. Wind energy

D. Geothermal energy

Show answer

Ans: Contested B or D — Nuclear **and** geothermal are both not stored solar. Wind and biomass are. Single-letter keys vary; learn the **pair**, not a coin-toss.

Confused pairs (minerals). German silver is a copper–nickel–zinc alloy and contains no silver metal. The **Mesabi** range is an **iron** story, not copper. The **Saar** is a **coal** field, not an iron-field dump. The **Peace Pipeline** name is **Iran–Pakistan**, not a Russia–Europe gas line. **Brent** crude is a North Sea class with **low** sulphur, not "zero sulphur".

N.9 GC extras — silver German · oil · uranium city

Note	Fact
German silver	Cu–Ni–Zn alloy — **no silver** (name from appearance)
Silver volume	Often **Mexico** #1 in recent frames
Aluminium ore	**Bauxite** (≠ haematite / magnetite / limonite)
Primary Al volume	Often **China** first in recent frames
Tin	Mostly **placer**; **Kinta Valley** (Malaysia) = tin
Brent crude	Major crude class; sourced from **North Sea**; low sulphur — not "zero sulphur"
Gasohol	~90% gasoline + ~10% ethanol; USA classic producer/consumer
Peace Pipeline	**Iran–Pakistan** gas pipeline name
First commercial oil (1857 register)	**Romania** (Drake 1859 Titusville is the other classic "first well" story)
Crude reserves	Often **Venezuela** #1 proved in IMYB frames; Saudi #2
SE Asia oil	**Indonesia**
Venezuela	Oil-dominated export economy
Ras Tanura	Refinery — **Saudi Arabia**
Pegu Yoma (Myanmar)	**Mineral oil** pair
Uranium City	**Canada** (Athabasca)
U production	Often **Kazakhstan** #1; reserves **Australia** (Olympic Dam)
Radium	From **pitchblende** (uranium ore)
Rare earths	**China** dominates; export curbs + electronics demand
Lithium Afghanistan find	Compared with oil discovery in Iran

Note	Fact
Chile	**Nitrate / iodine** (Atacama caliche)
Diamond volume	Often **Russia**
Cobalt (~¾ of batteries)	**DRC**
Camphor	Tree native to **China/Japan** (*Cinnamomum camphora*)

Non-ferrous metals. Coaching lists aluminium, copper, zinc, nickel, and tin as the main non-ferrous set. Pig iron and carbon steel are ferrous products and do not belong in that list.